

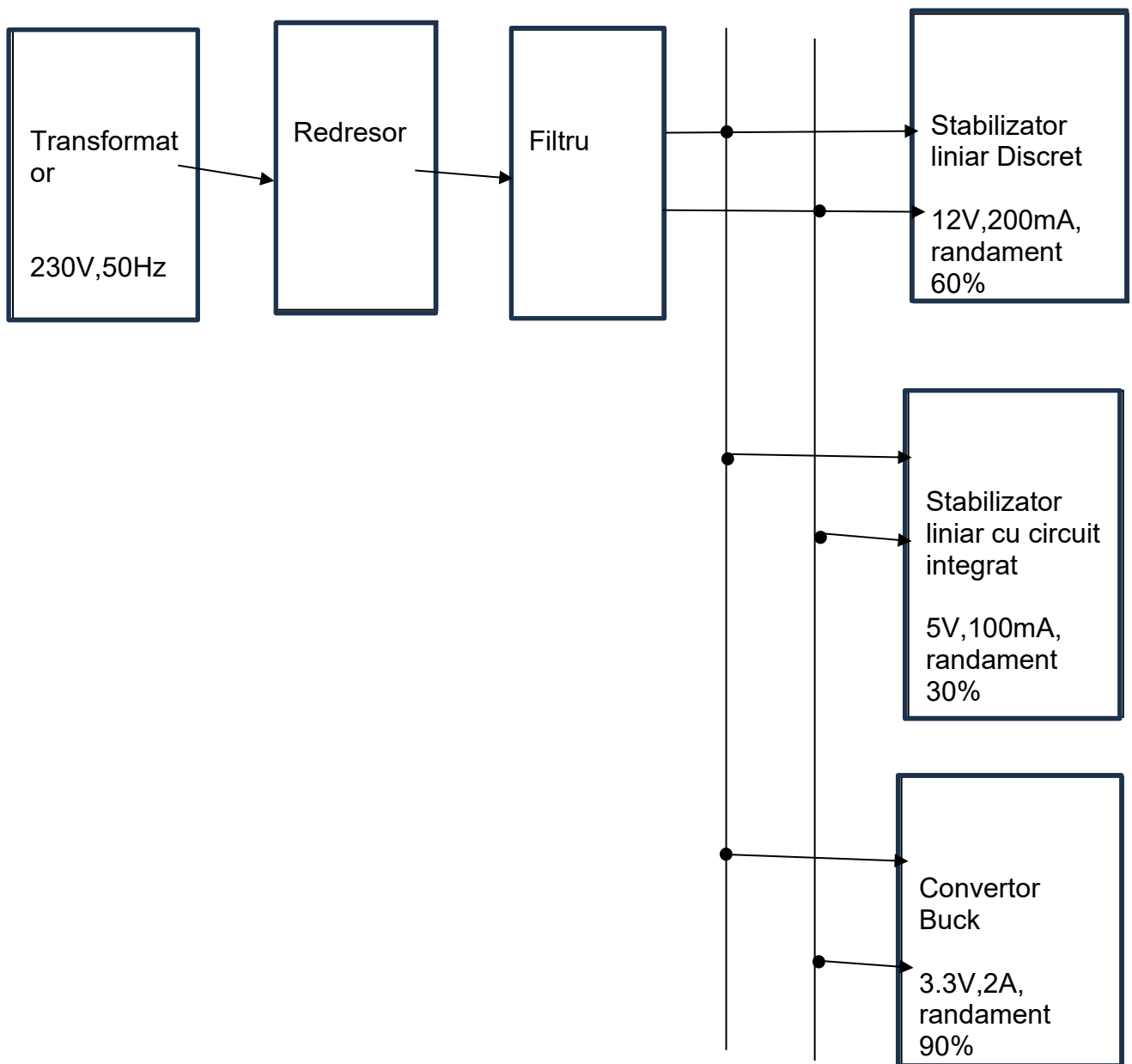
Proiect Surse de Alimentare

Rîmbulea Robert, grupa 2141, sg2

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1.Schema bloc



1.2.Calcul

Stiind formula clasica $P = V * I$, aflam puterea de iesire a fiecarui stabilizator:

1)Primul stabilizator: $P_{out_1} = V_{out} * I_{out} = 12V * 0.2A = 2.4W$

2)Al doilea stabilizator: $P_{out_2} = 5V * 0.1A = 0.5W$

3)Al treilea stabilizator: $P_{out_3} = 3.3V * 2A = 6.6W$

In continuare calculam puterea la intrarea fiecarui stabilizator folosind formula randamentului:

$$\eta = \frac{P_{out}}{P_{in}}$$

1)Primul stabilizator: $P_{in} = \frac{P_{out}}{\eta} = \frac{2.4W}{0.6} = 4W$

2)Al doilea stabilizator: $P_{in} = \frac{P_{out}}{\eta} = \frac{0.5W}{0.3} = 1.66W$

3)Al treilea stabilizator: $P_{in} = \frac{P_{out}}{\eta} = \frac{6.6W}{0.9} = 7.33W$

In continuare calcula V_{in} pt fiecare stabilizator:

1) $V_{in_1} = \frac{V_1}{\eta} = \frac{12V}{0.6} = 20V$

2) $V_{in_2} = \frac{V_2}{\eta} = \frac{5V}{0.3} = 16.66V$

3) $V_{in_3} = \frac{V_3}{\eta} = \frac{3.3V}{0.9} = 3.66V$

$$V_{inmax} = (V_{in1}, V_{in2})$$

$$V_t \in (13V, 16.6V) \Rightarrow \text{Aleg } V_t = 14V$$

In continuare calculam curentul de la intrarea fiecarui stabilizator folosind urmatoarea formula:

$$I_{in} = \frac{I_{out}}{\eta}$$

1)Primul stabilizator: $I_{in_1} = \frac{0.2A}{0.6} = 0.33A$

2)Al doilea stabilizator: $I_{in_2} = \frac{0.1A}{0.3} = 0.33A$

3)Al treilea stabilizator: $I_{in_3} = \frac{2}{0.9} = 2.2A$

$$\text{Din aceste calcul} \Rightarrow I_T = I_{in_1} + I_{in_2} + I_{in_3} = 0.33A + 0.33A + 2.2A = 2.86A$$

In continuare aflam V_t :

$$\text{La primul stabilizator avem } V_{out} = 12V \Rightarrow \text{Aleg } V_{in_{min}} = 13V$$

La al doilea stabilizator aflam valoarea lui V_{in_max} :

$$V_{in_max} = \frac{V_{out}}{n} = \frac{5V}{0.3} = 16.6$$

De aici rezulta ca: $V_t \in (V_{in_min}; V_{in_max}) \Rightarrow V_t \in (13V; 16.6V)$

2.Redresor si Transformator

Am ales sa folosesc dioda 1N5400 $\Rightarrow V_f = 1.2V$

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	3.0 A
V_{RRM}	50 V, 100 V, 200 V, 300 V, 500 V, 600 V, 800 V, 1000 V
I_{FSM}	200 A
I_R	5.0 μ A
V_f	1.2 V
T_J max.	150 °C
Package	DO-201AD
Diode variations	Single die

Stiind ca am $V_{tmin} = 13V$, $V_{tmax} = 16.6V$ calculez intervalul V_{rms} folosind urmatoarele formule: $V_{secrms1} = \frac{V_{tmin} + 2 \cdot V_f}{\sqrt{2}} = 10.88V$, si $V_{secrms2} = \frac{V_{tmax} + 2 \cdot V_f}{\sqrt{2}} = 13.43V$

B4	:	$\times$	$\checkmark$	f_x	$\checkmark$	$= (B2 + 2 \cdot B1) / \text{SQRT}(2)$
	A	B	C	D	E	F
1	Vf=	1.2				
2	Vtmin=	13				
3	Vtmax=	16.6				
4	Vsecrms1	10.88944				
B5	:	$\times$	$\checkmark$	f_x	$\checkmark$	$= (B3 + 2 \cdot B1) / \text{SQRT}(2)$
	A	B	C	D	E	F
1	Vf=	1.2				
2	Vtmin=	13				
3	Vtmax=	16.6				
4	Vsecrms1	10.88944				
5	Vsecrms2	13.43503				
6	Vrms=	12				

Eu am ales sa lucrez cu $V_{RMS} = 12V$

Aflam $V_{secmax} = V_{Rms} * \sqrt{2} = 16.97V$:

B7		⌵	:	✕	✓	<i>f_x</i>	⌵	=B6*SQRT(2)
	A	B	C	D	E			
1	Vf=	1.2						
2	Vtmin=	13						
3	Vtmax=	16.6						
4	Vsecrms1	10.88944						
5	Vsecrms2	13.43503						
6	Vrms=	12						
7	Vsecmax=	16.97056						

In continuare voi afla: $V_{dcmax} = V_{secmax} - 2 * V_f = 14.57V$

B8		⌵	:	✕	✓	f_x	⌵	=B7-2*B1
	A	B	C	D				
1	Vf=	1.2						
2	Vtmin=	13						
3	Vtmax=	16.6						
4	Vsecrms1	10.88944						
5	Vsecrms2	13.43503						
6	Vrms=	12						
7	Vsecmax=	16.97056						
8	Vdcmax=	14.57056						

Am ales $V_{riplu} = 2V$ si $f_{riplu} = 100Hz$

In continuare calculez: $V_{dcmin} = V_{dcmax} - V_{riplu} = 12.57V$:

B11	▼	:	✕	✓	f_x ▼	=B8-B9
	A	B	C	D		
1	Vf=	1.2				
2	Vtmin=	13				
3	Vtmax=	16.6				
4	Vsecrms1	10.88944				
5	Vsecrms2	13.43503				
6	Vrms=	12				
7	Vsecmax=	16.97056				
8	Vdcmax=	14.57056				
9	Vriplu=	2				
10	friplu=	100				
11	Vdcmin=	12.57056				

Aleg un nou $V_t = 14V$,deci recalculam I_t :

$$I_{in1} = \frac{V_1 * I_1}{V_t * n_1} = 0.28A$$

B16						
= (B13*B14)/(B12*B15)						
	A	B	C	D	E	F
1	Vf=	1.2				
2	Vtmin=	13				
3	Vtmax=	16.6				
4	Vsecrms1	10.88944				
5	Vsecrms2	13.43503				
6	Vrms=	12				
7	Vsecmax=	16.97056				
8	Vdcmax=	14.57056				
9	Vriplu=	2				
10	friplu=	100				
11	Vdcmin=	12.57056				
12	Vt=	14				
13	V1=	12				
14	I1=	0.2				
15	n1=	0.6				
16	Iin1=	0.285714				
17						

$$I_{in2} = \frac{V_2 * I_2}{V_t * n_2} = 0.11A$$

B20						
= (B17*B18)/(B12*B19)						
	A	B	C	D	E	F
1	Vf=	1.2				
2	Vtmin=	13				
3	Vtmax=	16.6				
4	Vsecrms1	10.88944				
5	Vsecrms2	13.43503				
6	Vrms=	12				
7	Vsecmax=	16.97056				
8	Vdcmax=	14.57056				
9	Vriplu=	2				
10	friplu=	100				
11	Vdcmin=	12.57056				
12	Vt=	14				
13	V1=	12				
14	I1=	0.2				
15	n1=	0.6				
16	Iin1=	0.285714				
17	V2=	5				
18	I2=	0.1				
19	n2=	0.3				
20	Iin2=	0.119048				

$$I_{in3} = \frac{V_3 * I_3}{V_t * n_3} = 0.52A$$

B24		: X ✓ f _x		=(B21*B22)/(B12*B23)		
	A	B	C	D	E	F
3	Vtmax=	16.6				
4	Vsecrms1	10.88944				
5	Vsecrms2	13.43503				
6	Vrms=	12				
7	Vsecmax=	16.97056				
8	Vdcmax=	14.57056				
9	Vriplu=	2				
10	friplu=	100				
11	Vdcmin=	12.57056				
12	Vt=	14				
13	V1=	12				
14	I1=	0.2				
15	n1=	0.6				
16	Iin1=	0.285714				
17	V2=	5				
18	I2=	0.1				
19	n2=	0.3				
20	Iin2=	0.119048				
21	V3=	3.3				
22	I3=	2				
23	n3=	0.9				
24	Iin3=	0.52381				

$$I_t = I_{in1} + I_{in2} + I_{in3} = 0.92A$$

B25		: X ✓ f _x		=B16+B20+B24		
	A	B	C	D	E	F
4	Vsecrms1	10.88944				
5	Vsecrms2	13.43503				
6	Vrms=	12				
7	Vsecmax=	16.97056				
8	Vdcmax=	14.57056				
9	Vriplu=	2				
10	friplu=	100				
11	Vdcmin=	12.57056				
12	Vt=	14				
13	V1=	12				
14	I1=	0.2				
15	n1=	0.6				
16	Iin1=	0.285714				
17	V2=	5				
18	I2=	0.1				
19	n2=	0.3				
20	Iin2=	0.119048				
21	V3=	3.3				
22	I3=	2				
23	n3=	0.9				
24	Iin3=	0.52381				
25	I _t =	0.928571				

In continuare află valoarea condensatorului: $C = \frac{It}{f_{riplu} \cdot V_{riplu}} = \frac{0.9286A}{200} = 4643\mu F$

B26				
	A	B	C	D
5	Vsecrms2	13.43503		
6	Vrms=	12		
7	Vsecmax=	16.97056		
8	Vdcmax=	14.57056		
9	Vriplu=	2		
10	friplu=	100		
11	Vdcmin=	12.57056		
12	Vt=	14		
13	V1=	12		
14	I1=	0.2		
15	n1=	0.6		
16	Iin1=	0.285714		
17	V2=	5		
18	I2=	0.1		
19	n2=	0.3		
20	Iin2=	0.119048		
21	V3=	3.3		
22	I3=	2		
23	n3=	0.9		
24	Iin3=	0.52381		
25	It=	0.928571		
26	C=	0.004643		
27				

Aflăm valoarea rezistenței de sarcină: $R_s = \frac{V_t}{I_t} = \frac{14V}{0.9286A} = 15.07\Omega$

B27				
	A	B	C	D
6	Vrms=	12		
7	Vsecmax=	16.97056		
8	Vdcmax=	14.57056		
9	Vriplu=	2		
10	friplu=	100		
11	Vdcmin=	12.57056		
12	Vt=	14		
13	V1=	12		
14	I1=	0.2		
15	n1=	0.6		
16	Iin1=	0.285714		
17	V2=	5		
18	I2=	0.1		
19	n2=	0.3		
20	Iin2=	0.119048		
21	V3=	3.3		
22	I3=	2		
23	n3=	0.9		
24	Iin3=	0.52381		
25	It=	0.928571		
26	C=	0.004643		
27	Rs=	15.07692		

Aleg $N1 = 10$, si $N2$ are valoarea lui $V_{secmax} = 16.97V$

In continuare aflu raportul spirelor: $n = \frac{N1}{N2} = \frac{324}{18.38} = 19.09$

B30				
	A	B	C	D
10	friplu=	100		
11	Vdcmin=	12.57056		
12	Vt=	14		
13	V1=	12		
14	I1=	0.2		
15	n1=	0.6		
16	Iin1=	0.285714		
17	V2=	5		
18	I2=	0.1		
19	n2=	0.3		
20	Iin2=	0.119048		
21	V3=	3.3		
22	I3=	2		
23	n3=	0.9		
24	Iin3=	0.52381		
25	It=	0.928571		
26	C=	0.004643		
27	Rs=	15.07692		
28	N1=	324		
29	N2=	16.97056		
30	n=	19.09188		
31	n^2=	364.5		

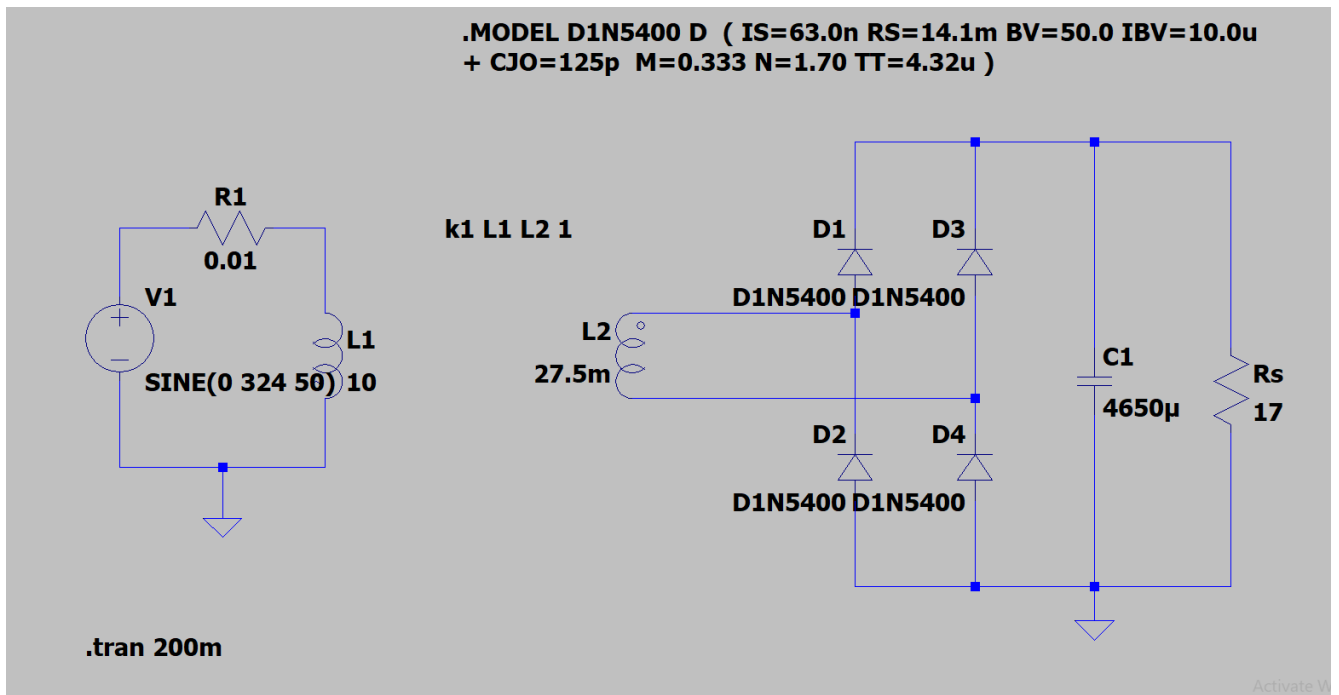
$$n^2 = 19.09^2 = 364,52$$

B31				
	A	B	C	D
10	friplu=	100		
11	Vdcmin=	12.57056		
12	Vt=	14		
13	V1=	12		
14	I1=	0.2		
15	n1=	0.6		
16	Iin1=	0.285714		
17	V2=	5		
18	I2=	0.1		
19	n2=	0.3		
20	Iin2=	0.119048		
21	V3=	3.3		
22	I3=	2		
23	n3=	0.9		
24	Iin3=	0.52381		
25	It=	0.928571		
26	C=	0.004643		
27	Rs=	15.07692		
28	N1=	324		
29	N2=	16.97056		
30	n=	19.09188		
31	n^2=	364.5		

Aleg $L1 = 10H \Rightarrow L2 = \frac{L1}{364,52} = 27mH$

B33	▼	:	✖	✓	fx	▼	=B32/B31
	A	B	C	D	E		
16	lin1=	0.285714					
17	V2=	5					
18	I2=	0.1					
19	n2=	0.3					
20	lin2=	0.119048					
21	V3=	3.3					
22	I3=	2					
23	n3=	0.9					
24	lin3=	0.52381					
25	It=	0.928571					
26	C=	0.004643					
27	Rs=	15.07692					
28	N1=	324					
29	N2=	16.97056					
30	n=	19.09188					
31	n^2=	364.5					
32	L1=	10					
33	L2=	0.027435					
34							
35							
36							

2.1.Circuit in Ltspice



Am modificat unele valori pt a avea rezultate mai bune:

$$L2 = 27mH \Rightarrow L2 = 27.5mH$$

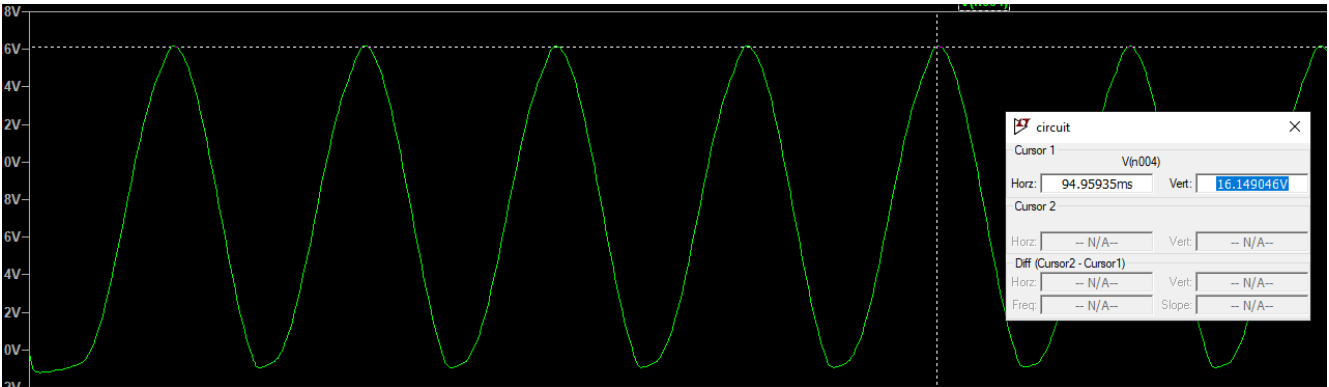
$$C = 4643\mu F \Rightarrow C = 4650\mu F$$

$$Rs = 15.07\Omega \Rightarrow Rs = 17\Omega$$

2.2.Simulari

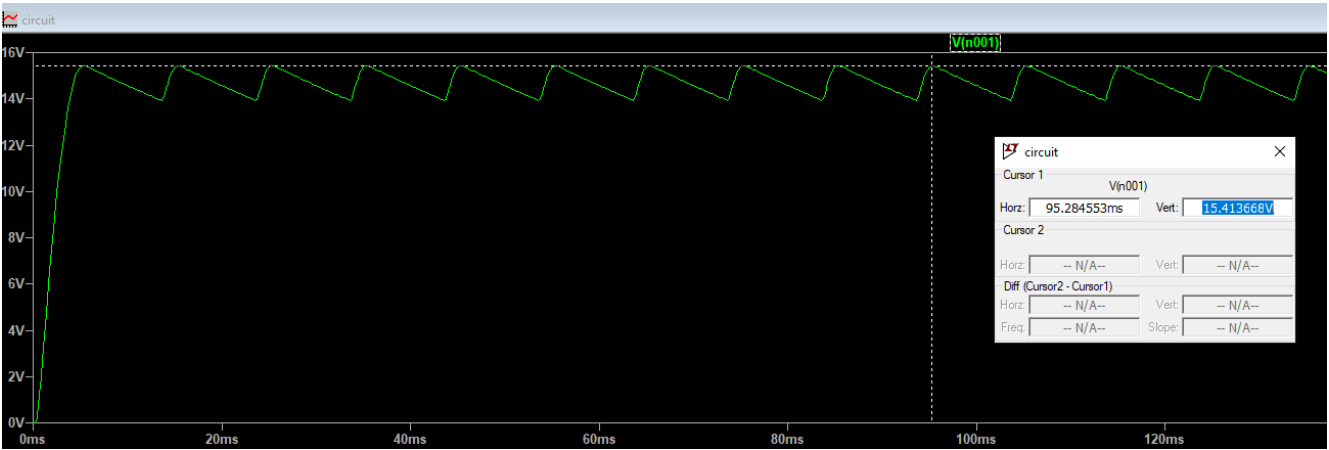
Vsecmax sim=16.14V

Vsecmax calc=16.97V



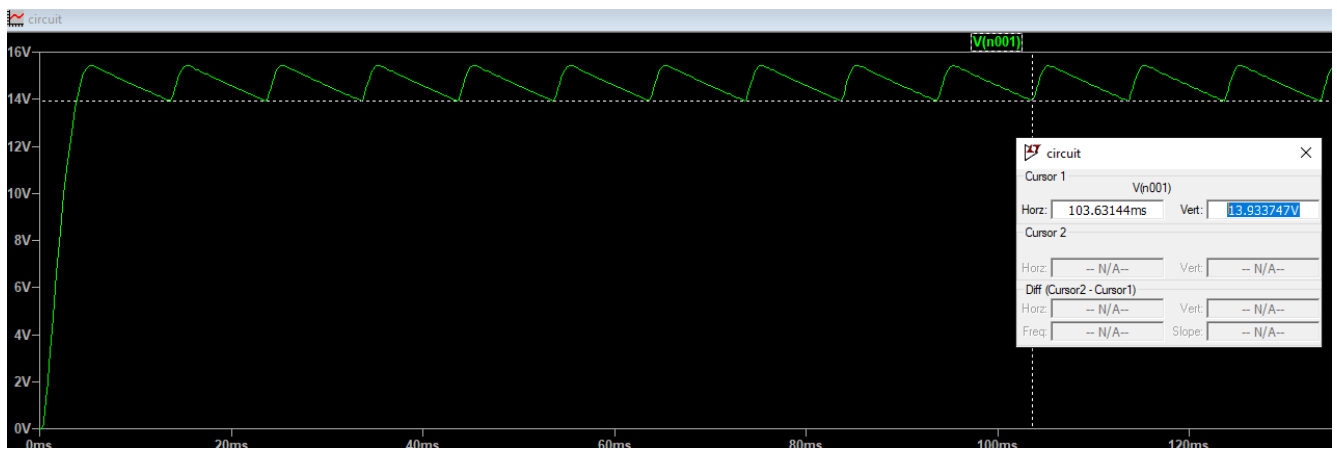
Vdcmaxsim=15.41V

Vdcmax calc=14.57

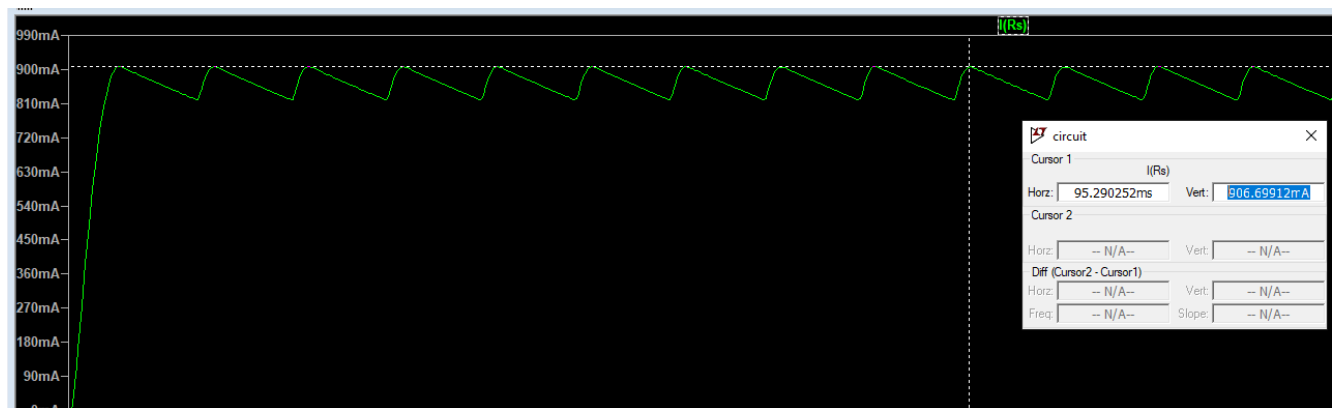


Vdcmin sim=13.93V

Vdcmin calc=12.57V



Curent masurat=906,69mA
 Curent Calculat=928.6mA



3.Stabilizator liniar

3.1.Alegerea diodei

Avem nevoie de o dioda care sa scoata $V_z - V_{BE} = 12V$.

Am ales sa folosesc dioda BZX55C13 care are urmatoarele caracterisitici:

- $V_{nominala} = 13V$

- $V \in [12.4V; 14.1V]$

- $i_{ZTmax} = 5mA$

-pret:1.5 lei

-datasheet:

Type	V_{Znom}	I_{ZT}	for V_{ZT} and
BZX55C...	V	mA	V ¹⁾
2V4	2.4	5	2.28 to 2.56
2V7	2.7	5	2.5 to 2.9
3V0	3.0	5	2.8 to 3.2
3V3	3.3	5	3.1 to 3.5
3V6	3.6	5	3.4 to 3.8
3V9	3.9	5	3.7 to 4.1
4V3	4.3	5	4.0 to 4.6
4V7	4.7	5	4.4 to 5.0
5V1	5.1	5	4.8 to 5.4
5V6	5.6	5	5.2 to 6.0
6V2	6.2	5	5.8 to 6.6
6V8	6.8	5	6.4 to 7.2
7V5	7.5	5	7.0 to 7.9
8V2	8.2	5	7.7 to 8.7
9V1	9.1	5	8.5 to 9.6
10	10	5	9.4 to 10.6
11	11	5	10.4 to 11.6
12	12	5	11.4 to 12.7
13	13	5	12.4 to 14.1

3.2.Alegerea tranzistorului

Am ales sa folosesc tranzistorul QBD135 care are urmatoarele specificatii:

-pret:3.25 lei

- $V_{BE} = 0.7V$

- $\beta = 40$

- $i_{Smin} = 100mA$

- $i_{Smax} = 200mA$

3.3.Calculare

-Aflam valoarea lui i_{zt_min} calculand 10% din cel maxim:

B2 $\downarrow$: $\times \checkmark f_x \downarrow$ $= (10/100) * (B1)$

	A	B	C	D	E
1	iztmax=	0.005			
2	iztmin=	0.0005			

Rezulta $i_{zt_min} = 0.5mA$

-In continuare aflam valoarea rezistentei:

$$R1_{max} = \frac{Vdc_{max} - V_z}{izt_{max} + \frac{is_{min}}{\beta + 1}} = 211\Omega$$

7 $\downarrow$: $\times \checkmark f_x \downarrow$ $= (B3 - B5) / (B1 + (B6 / (B4 + 1)))$

	A	B	C	D	E	F
	iztmax=	0.005				
	iztmin=	0.0005				
	Vdcmax=	14.57				
	β =	40				
	Vz=	13				
	ismin=	0.1				
	R1_max=	211.0492				

-la fel pentru $R1_{min}$:

$$R1_{min} = \frac{Vdc_{min} - V_z}{izt_{min} + \frac{is_{max}}{\beta + 1}} = -43.53\Omega,$$

B11 $\downarrow$: $\times \checkmark f_x \downarrow$ $= (B10 - B5) / (B1 + (B9 / (B4 + 1)))$

	A	B	C	D	E	F	G
1	iztmax=	0.005					
2	iztmin=	0.0005					
3	Vdcmax=	14.57					
4	β =	40					
5	Vz=	13					
6	ismin=	0.1					
7	R1_max=	211.0492					
8							
9	ismax=	0.2					
10	Vdcmin=	12.57					
11	R1_min=	-43.5309					
12							

diferenta dintre Vdcmin si Vz este -0.43,dar daca fac in modul rezultatul este=80.07 Ω .

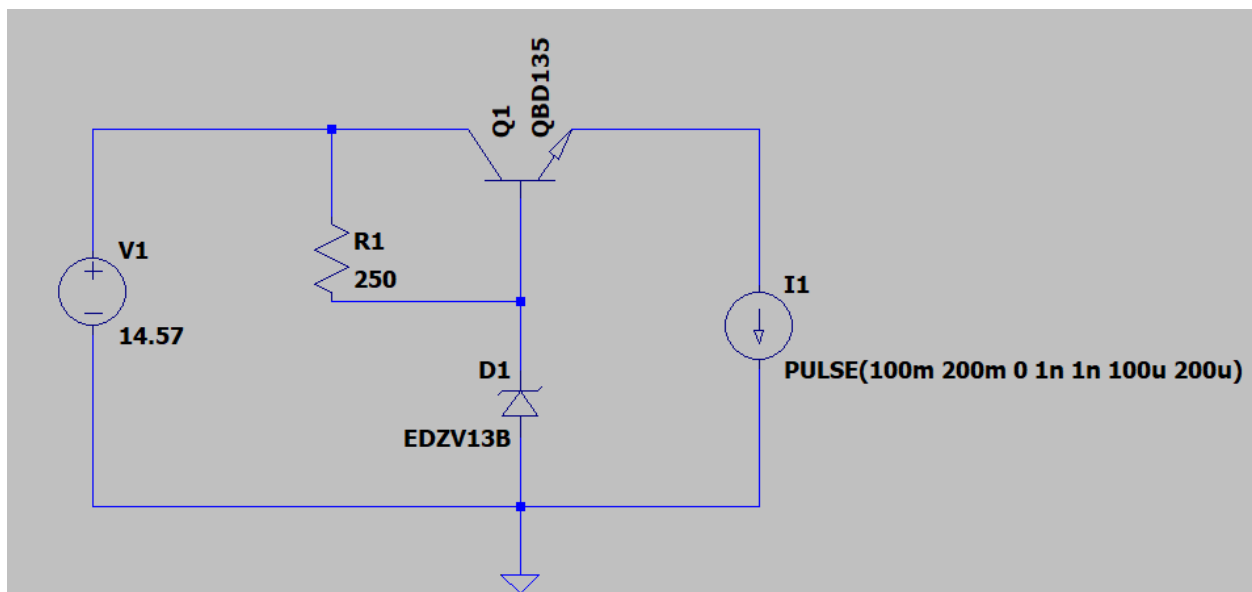
-in continuare aflam valoarea rezistentei de sarcina:

$$R_o = \frac{V_{termica}}{is_{max}} = 0.125\Omega$$

Luam valoarea tensiunii termice de 25mV.

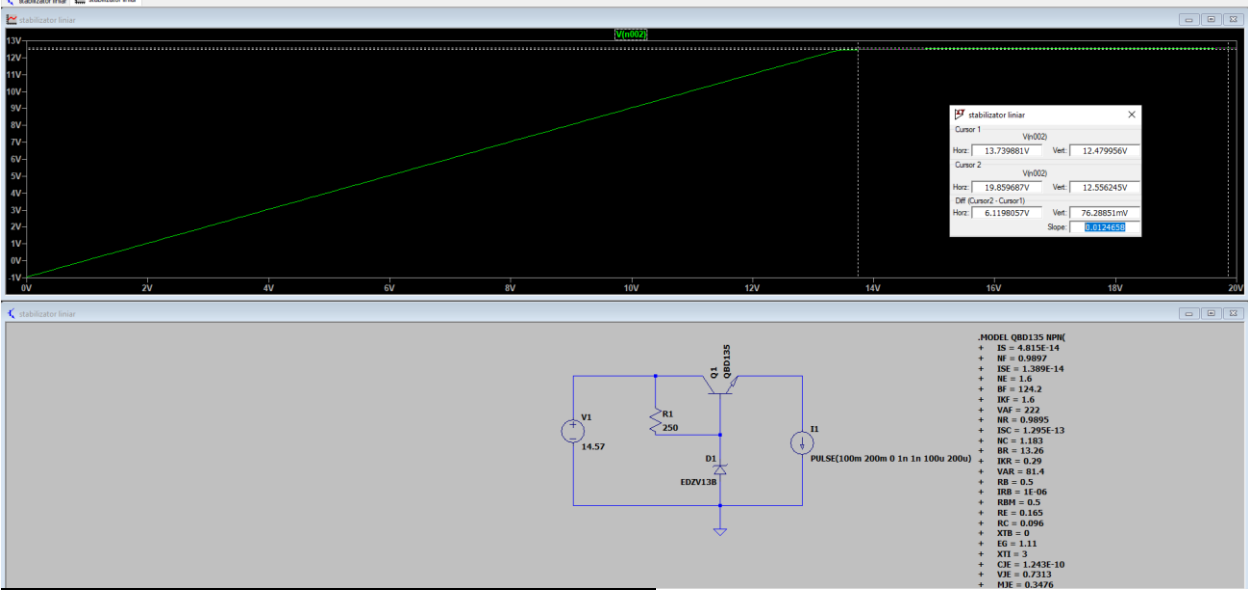
B15		:	$\times$	$\checkmark$	f_x	$\vee$	=B14/B9
	A	B	C	D			
1	iztmax=	0.005					
2	iztmin=	0.0005					
3	Vdcmax=	14.57					
4	β =	40					
5	Vz=	13					
6	ismin=	0.1					
7	R1_max=	211.0492					
8							
9	ismax=	0.2					
10	Vdcmin=	12.57					
11	R1=	-43.5309					
12							
13							
14	Vtermica=	0.025					
15	Ro=	0.125					
16							

3.4..Implementarea circuitului



3.5.Simulari

-Simulare SUI:

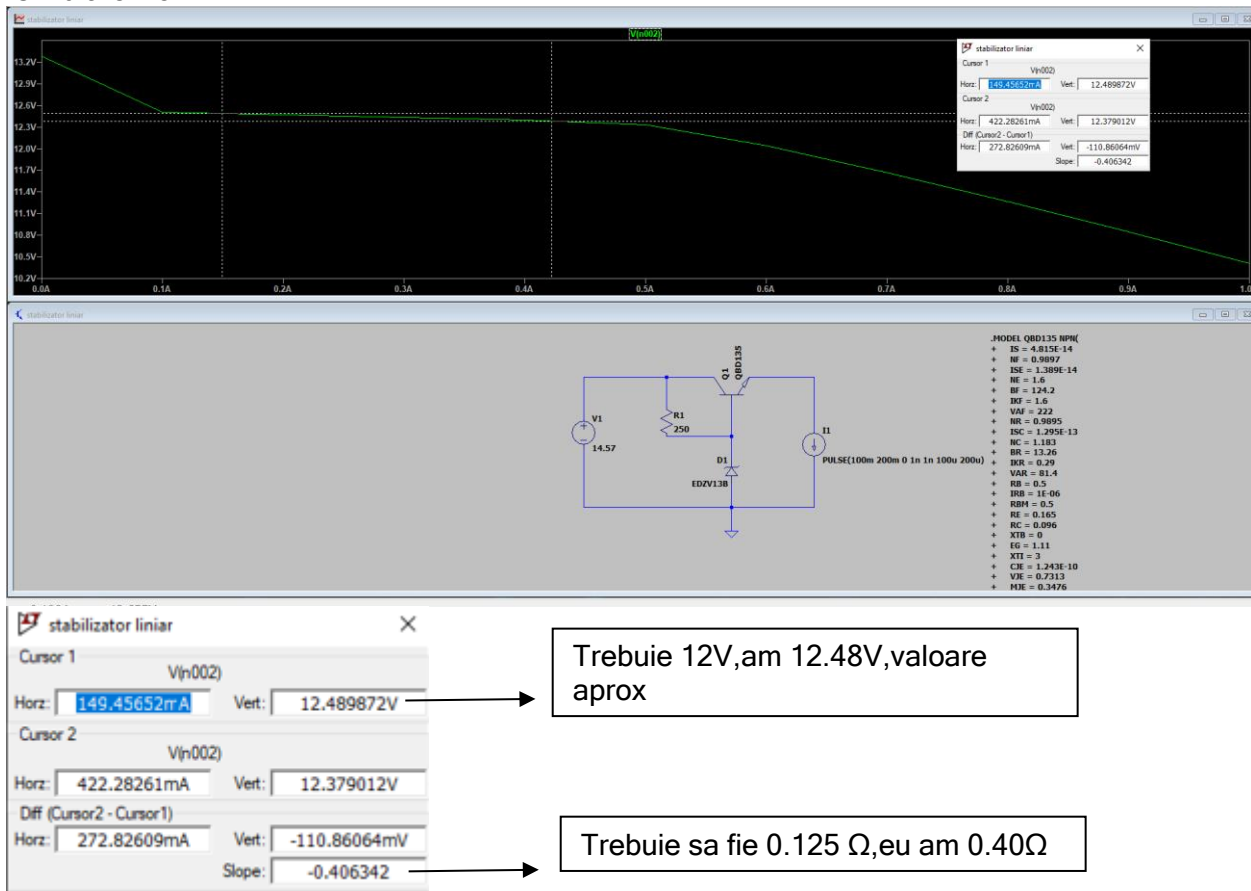


Facem raportul $\frac{1}{0.0124658} = 80.21948$,care este mai mare decat 80

=1/F1

D	E	F	G
	slope=	0.012466	
	Sui=	80.21948	

-simulare Ro:



4.Stabilizator de tensiune cu circuit integrat

4.1.Alegerea stabilizatorului

Am ales sa folosesc stabilizatorul **LM7805** deoarece:

- oferă o tensiune de ieșire fixă de 5V
- este simplu de utilizat datorită celor trei pini
- ușor de gasit
- accesibil ca preț:2,07 RON

4.2.Calcul

Date:-Vin=14.57V

-Vout=5V

-Iout=0.1A

Stiind aceste date,pot calcula puterea disipata:

$$P = (V_{in} - V_{out}) * I_{out} = (14.57V - 5V) * 0.1A = 0.957W$$

B5 : $\times$ $\checkmark$ f_x $\downarrow$ =(B1-B2)*B3					
	A	B	C	D	E
1	Vin=	14.57			
2	Vout=	5			
3	Iout=	0.1			
4					
5	P=	0.957			
6					

Aleg $T_A = 30^\circ C$,si $T_{max} = 125^\circ C$

Am valoarea rezistentei termice de 65Ω

Symbol	Parameter		Value
V _I	Input Voltage	V _O = 5 V to 18 V	35
		V _O = 24 V	40
R _{θJC}	Thermal Resistance, Junction-Case (TO-220)		5
R _{θJA}	Thermal Resistance, Junction-Air (TO-220)		65

In continuare voi calcula temperatura pentru a vedea daca este necesara utilizarea unui radiator:

$$T_J = T_A + (R_{ja} * P) = 30^\circ C + (65\Omega * 0.957W) = 92.20^\circ C$$

=F1+(F3*B5)				
D	E	F	G	H
	Ta=	30		
	Tmax=	125		
	Rja=	65		
	Tj=	92.205		

Temperatura este mai mica decat $125^\circ C$,deci nu necesita neaparat radiator,dar tot il voi introduce.

Am ales sa folosesc **TO-220 Aluminum Heat Sink** deoarece:

- are dimensiuni compacte ușor de integrat
- este realizat din aluminiu cu bună conductivitate termică
- este accesibil ca si pret:9,55 RON(Ebay)

În continuare voi calcula rezistența de sarcină:

$$R_s = \frac{V_{out}}{I_{out}} = \frac{5V}{0.1A} = 50\Omega$$

B6					=B2/B3				
	A	B	C	D					
1	Vin=	14.57							
2	Vout=	5							
3	Iout=	0.1							
4									
5	P=	0.957							
6	Rs=	50							

Conform datasheet-ului, trebuie să mai adăugăm în circuit un condensator de 0,33 μF, și unul de 0,1 μF.

C 33uF/63Vdc 105grdC 6.3x11mm Elite

Cod produs: 107063



1,00 lei /Buc

Status: 90 în stoc

CONDENSATOR ELECTROLITIC RADIAL 33UF 63VDC 20% 105GRDC 6.3X11MM R2.5MM

Producator: Elite

Temperatura: -40...+105grdC

Tensiune: 63Vdc

Capacitanta (F): 33uF

Cantitate

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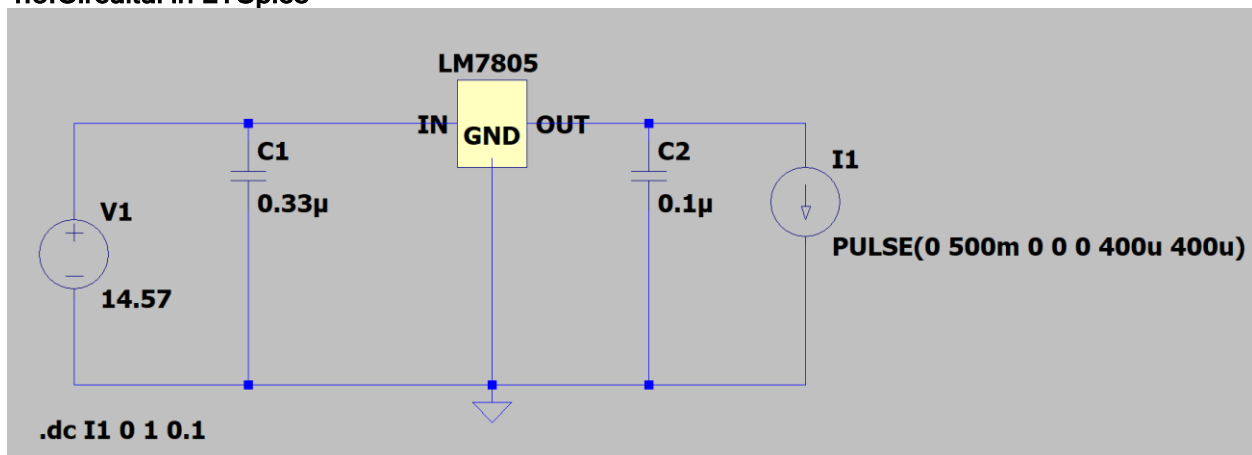
Highlights

- 50V
- Non-polaris
- 0.1uF
- Original, Hig

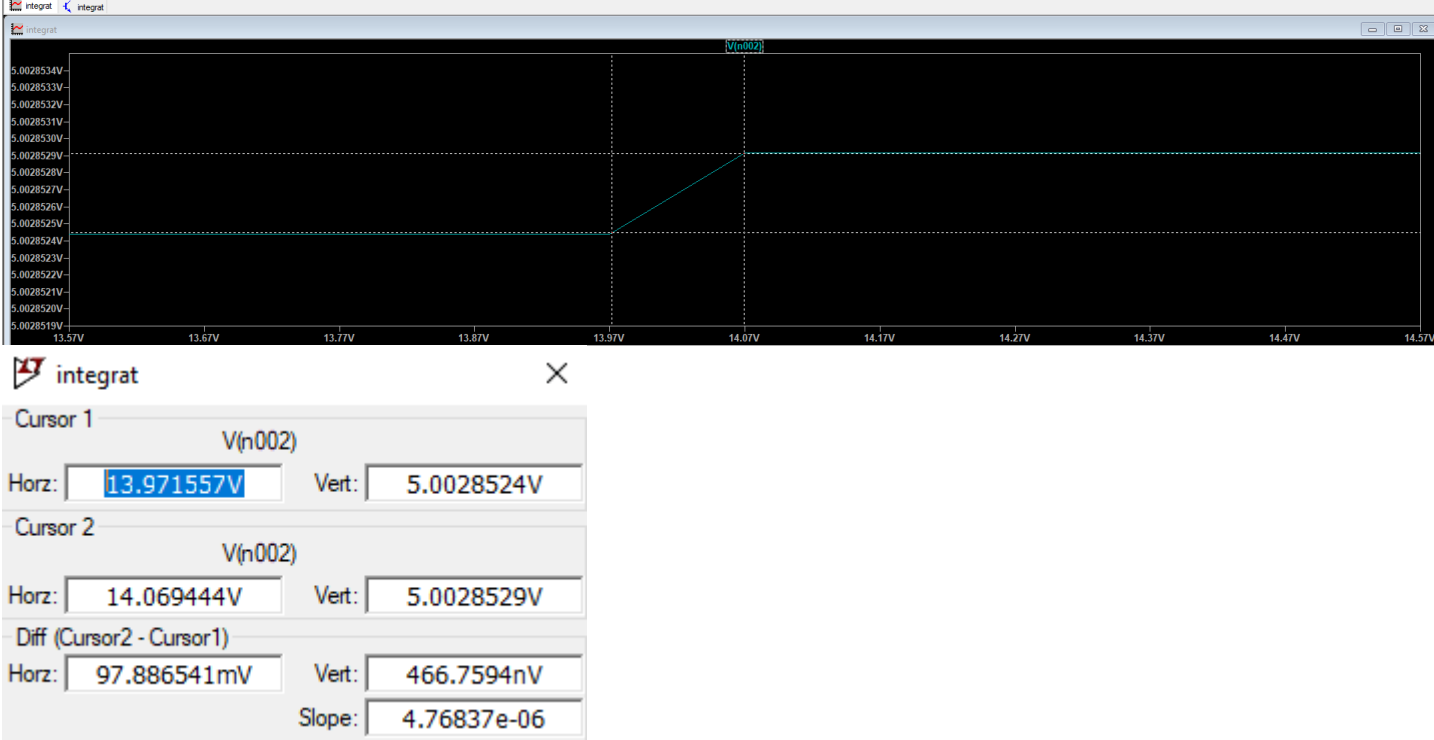
Services

- Cash on

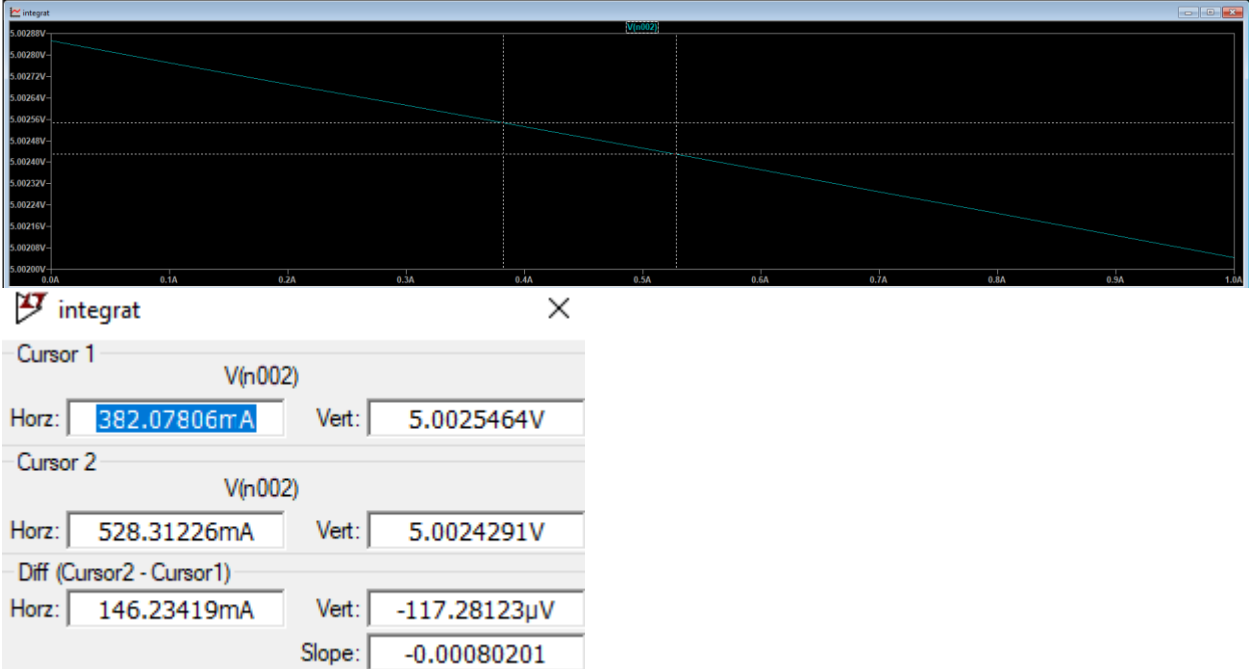
4.3.Circuitul in LTSpice



4.4.Simulari
-SUI:



-rezistenta de iesire:



5. Convertor Buck

5.1. Specificatii:

- $V_{out}=3.3V$

- $I_o=2A$

- $\eta \geq 90\%$

- $V_{dfr}=0.6V$

- $P_o = (V_o + V_{dfr}) * I_o = (3.3V + 0.6V) * 2A = 7.8A$

B6		:	$\times$	$\checkmark$	f_x	$\checkmark$	$= (B1+B4)*B2$
	A	B	C	D	E		
1	Vout=	3.3					
2	Io=	2					
3	n=	0.9					
4	Vdfr=	0.6					
5							
6	Po=	7.8					

- $V_{in_{min}} = 13.3V$

- $V_{in_{max}} = 15.3V$

- $I_{in1} = 440 mA$

- $I_{in2} = 560 mA$

- $V_p=0.2 V$

- $P_{in} = 7.33W$

- $V_{in_{nom}} = \frac{V_{in_{min}}+V_{in_{max}}}{2} = \frac{13.3V+15.3V}{2} = 14.3V$

$\checkmark$	$= (E1+E2)/2$				
	D	E	F		
	Vin_min=	13.3			
	Vin_max=	15.3			
	Pin=	7.33			
	Iin_1=	0.44			
	Iin_2=	0.56			
	Vin_nom=	14.3			

- $V_{oi} = V_o + V_{dfr} + V_p = 3.3V + 0.6V + 0.2 = 4.1V$

	$=B1+B4+H1$				
	D	E	F	G	H
	Vin_min=	13.3		Vp=	0.2
	Vin_max=	15.3		Voi=	4.1
	Pin=	7.33			

- $f_{sw} = 200 kHz \Rightarrow T_{sw} = \frac{1}{f_{sw}} = \frac{1}{200 kHz} = 5\mu s$

B11 : $\times$ $\checkmark$ f_x =1/B10				
	A	B	C	D
1	Vout=	3.3		Vin_min=
2	Io=	2		Vin_max=
3	n=	0.9		Pin=
4	Vdfw=	0.6		lin_1=
5				lin_2=
6	Po=	7.8		
7				Vin_nom=
8				
9				
10	fsw=	200000		
11	Tsw=	0.000005		

5.2.Dimensionari

-factor de umplere maxim:

$$D_{max} = \frac{V_{oi}}{V_{inmax}} = \frac{4.1V}{15.3V} = 0.26$$

$$T_{on_{max}} = T_{sw} * D_{max} = 5\mu s * 0.26 = 1.33\mu s$$

B14 : $\times$ $\checkmark$ f_x =B13*B11				
	A	B	C	D
1	Vout=	3.3		Vin_min=
2	Io=	2		Vin_max=
3	n=	0.9		Pin=
4	Vdfw=	0.6		lin_1=
5				lin_2=
6	Po=	7.8		
7				Vin_nom=
8				
9				
10	fsw=	200000		
11	Tsw=	0.000005		
12				
13	Dmax=	0.267973856		
14	Ton_max=	1.33987E-06		

-factor de umplere minim

$$D_{min} = \frac{V_{oi}}{V_{inmin}} = \frac{4.1V}{13.3V} = 0.30$$

$$T_{on_{min}} = T_{sw} * D_{min} = 5\mu s * 0.30 = 1.54\mu s$$

B14 : $\times$ $\checkmark$ f_x =B11*B13				
	A	B	C	D
1	Vout=	3.3		Vin_min=
2	Io=	2		Vin_max=
3	n=	0.9		Pin=
4	Vdfw=	0.6		lin_1=
5				lin_2=
6	Po=	7.8		
7				Vin_nom=
8				
9				
10	fsw=	200000		
11	Tsw=	0.000005		
12				
13	Dmin=	0.308270677		
14	Ton_min=	1.54135E-06		

-factor de umplere la tensiune nominala :

$$Ton_{max} = T_{sw} * D_{max} = 5\mu s * 0.28 = 1.4\mu s$$

Se alege dioda STPS15L30CB:



Symbol	Value
$I_{F(AV)}$	2 x 7.5 A
V_{RRM}	30 V
T_j	150 °C
V_F (typ)	0.34 V

$$I_L = \frac{20}{100} * I_O = 0.2 * 2A = 0.4A$$

$$=(20/100)*B2$$

	D	E	F	G	H	I	J	K	L
	Vin_min=	13.3		Vp=	0.2		Vrp=	0.05	
	Vin_max=	15.3		Voi=	4.1		Il=	0.4	

$$V_f = \left(\frac{V_{oi}}{T_{on_{min}}} \right) * T_{sw} = \left(\frac{4.1}{1.54\mu s} \right) * 5\mu s = 13.3$$

=(H2/B17)*B11

	D	E	F	G
	Vin_min=	13.3		Vp=
	Vin_max=	15.3		Voi=
	Pin=	7.33		
	Iin_1=	0.44		
	Iin_2=	0.56		
	Vin_nom=	14.3		
			VF=	13.3

$$L = \frac{((V_f - V_o) * T_{on_{max}})}{2 * I_o} = \frac{(13.3 - 3.3) * 1.33\mu s}{2 * 2} = 3.34\mu H$$

$$C = \frac{1}{0.25 * V_{rp} * 8 * f_{sw}} = \frac{1}{0.25 * 50mV * 8 * 200 kHz} = 50\mu F$$

$$=1/(0.25*K1*8*B10)$$

	D	E	F	G
	Vin_min=	13.3		Vp=
	Vin_max=	15.3		Voi=
	Pin=	7.33		
	Iin_1=	0.44		
	Iin_2=	0.56		
	Vin_nom=	14.3		
			VF=	13.3
			L=	3.34967E-06
			C=	0.00005

$$=((G10-B1)*B14)/4$$

	D	E	F	G
	Vin_min=	13.3		Vp=
	Vin_max=	15.3		Voi=
	Pin=	7.33		
	Iin_1=	0.44		
	Iin_2=	0.56		
	Vin_nom=	14.3		
			VF=	13.3
			L=	3.34967E-06

Condensator electrolitic THT
4.7uF 100Vdc

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100 lei

SKU: CNDNSTR-01665

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Multi H Q, Ind 3.3uh Q35 50ma 2pads Smt

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Mfr: [Eaton](#)

RF Inductors - SMD Multi H Q, IND 3.3uH Q35 50mA 2PADS SMT

Datasheet: [PDF Datasheet](#)

\$0.10




5.4. Tranzistorului MOSFET



Am ales să folosesc tranzistorul TN2640:

-pret 4.88 RON

- $R_{DSon} = 3.5 \Omega$

- $R_{DSoff} = 3.5 \Omega$

- $V_{GS} = 2V$

-aleg $V_{com} = 12V$ și $I_{com} = 5A$

Pierderile în conductie: P_{cond}

$$P_{cond} = R_{DSon} * I_O^2 * D_{max} = 10.6m\Omega * 4 * 0.26 = 0.011W$$

=L6*4*B13

	D	E	F	G	H	I	J	K	L
Vin_min=	13.3		Vp=	0.2		Vrp=	0.05		
Vin_max=	15.3		Voi=	4.1		IL=	0.4		
Pin=	7.33								
lin_1=	0.44								
lin_2=	0.56								
								Rds_on=	0.0106
Vin_nom=	14.3							Pcond=	0.011362

-pierderi in comutatie:

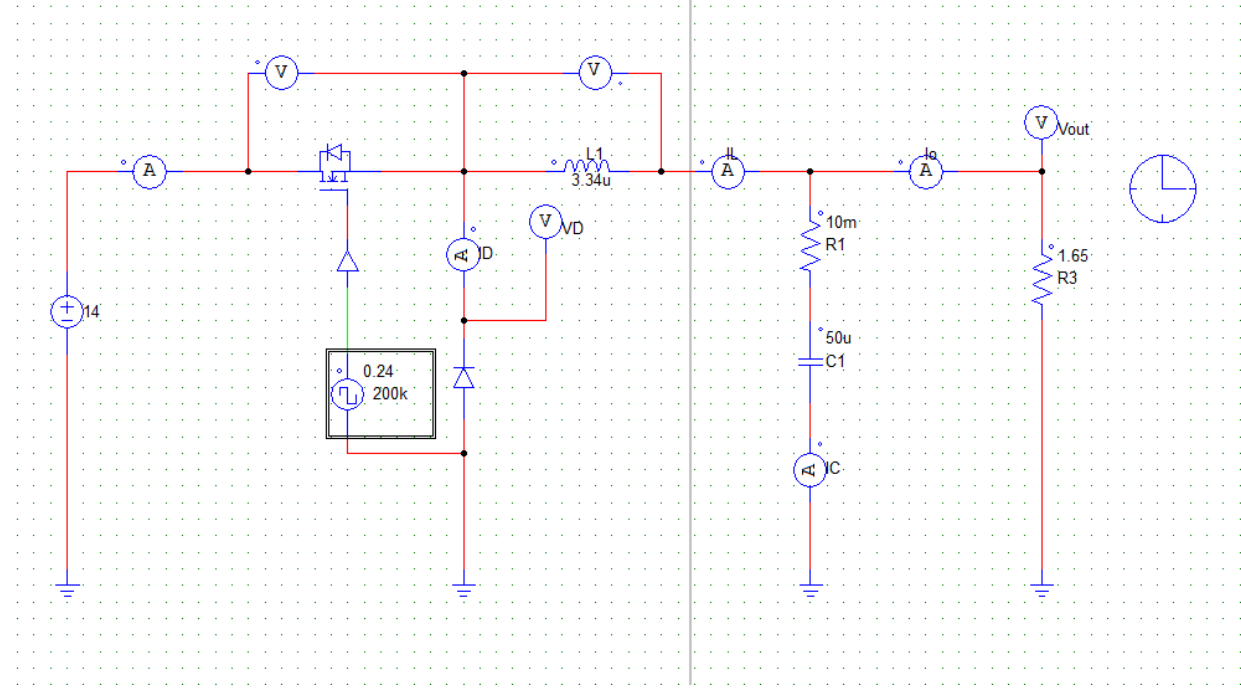
$$Icom_{LH} = \frac{V_{com}-V_{gs}}{R_{dson}} = \frac{12V-2V}{3.5\Omega} = 2.85A$$

$$Icom_{HL} = \frac{V_{com}-V_{gs}}{R_{dsoff}} = \frac{12V-2V}{3.5\Omega} = 2.85A$$

=(01-02)/04

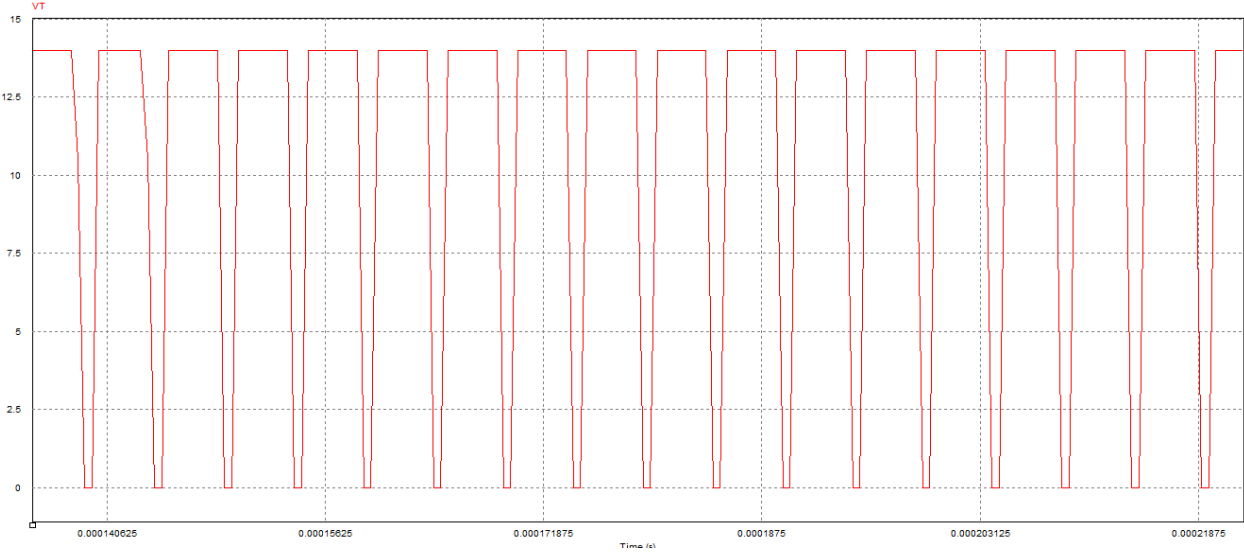
	D	E	F	G	H	I	J	K	L	M	N	O
Vin_min=	13.3		Vp=	0.2		Vrp=	0.05			Vcom=	12	
Vin_max=	15.3		Voi=	4.1		IL=	0.4			Vgs=	2	
Pin=	7.33									Rdson=	3.5	
lin_1=	0.44									Rdsoff=	3.5	
lin_2=	0.56											
								Rds_on=	0.0106		Icom_LH=	2.857143
Vin_nom=	14.3							Pcond=	0.011362		Icom_HL=	2.857143

5.5.Circuit in PSIM

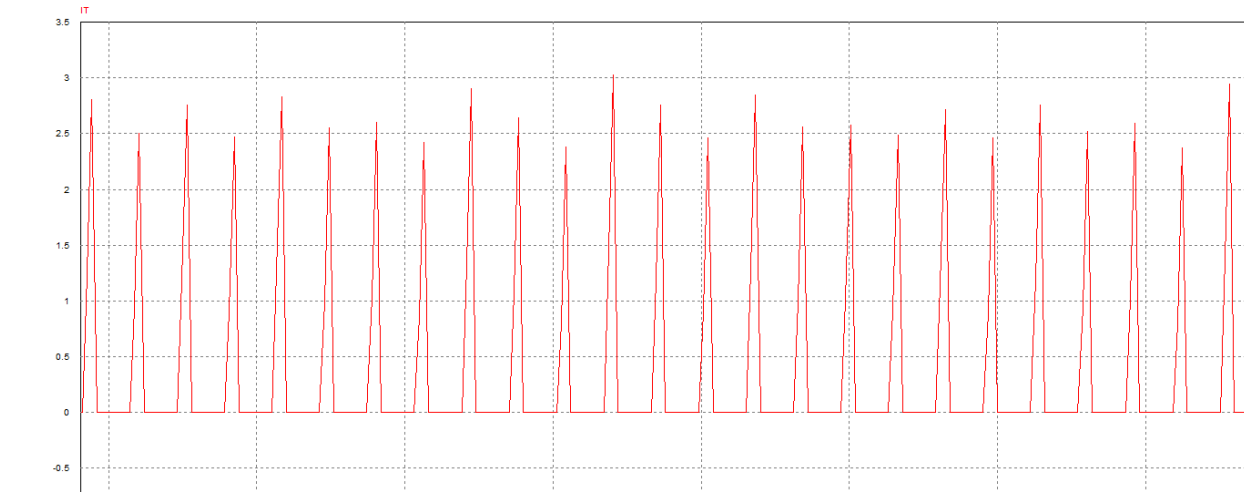


5.6.Simulari in PSIM

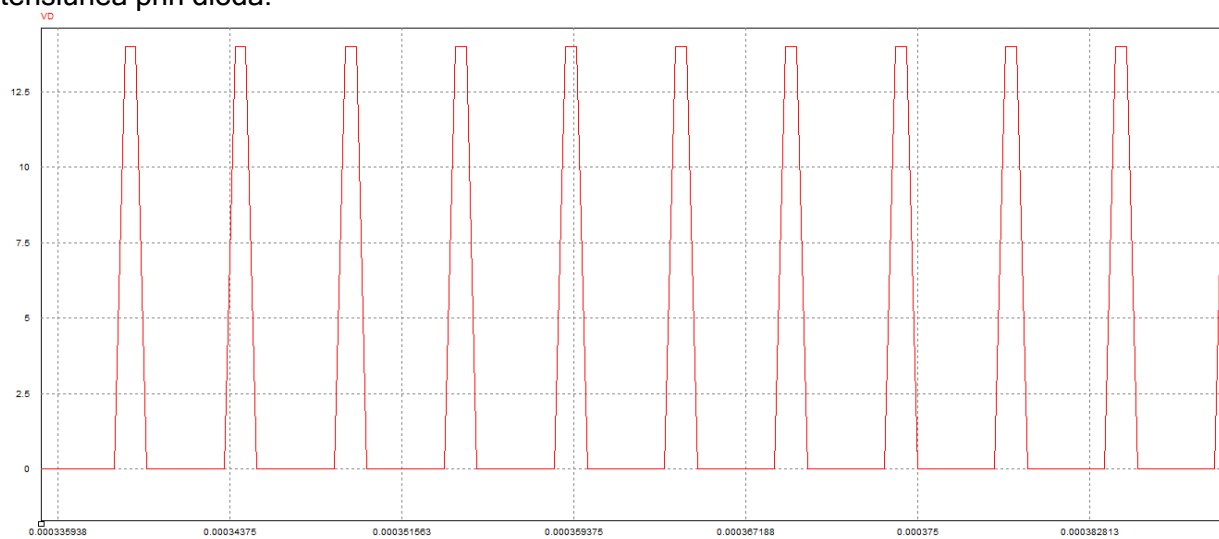
-tensiunea prin tranzistor:



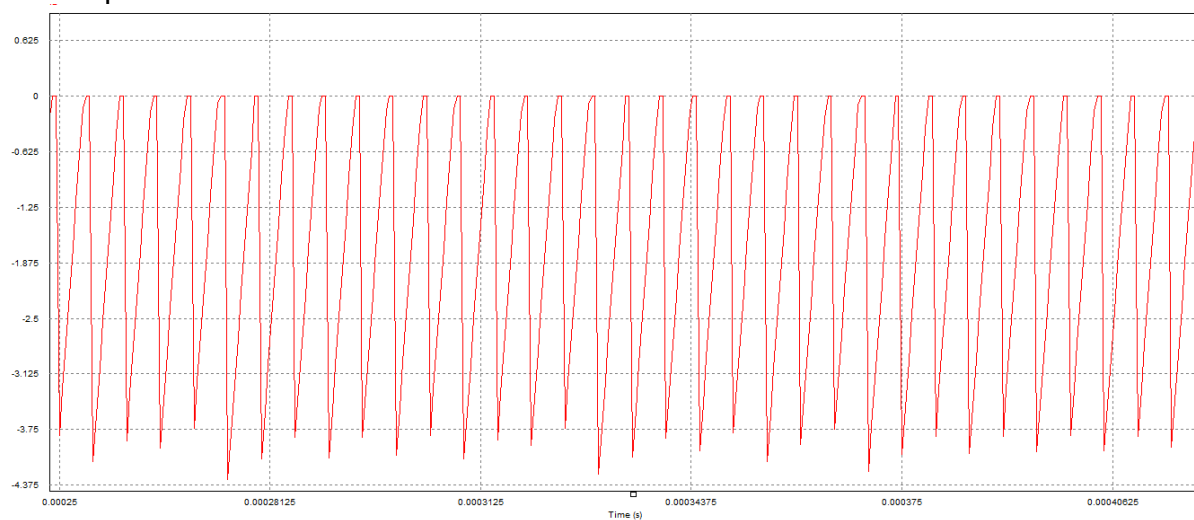
-curentul prin tranzistor:



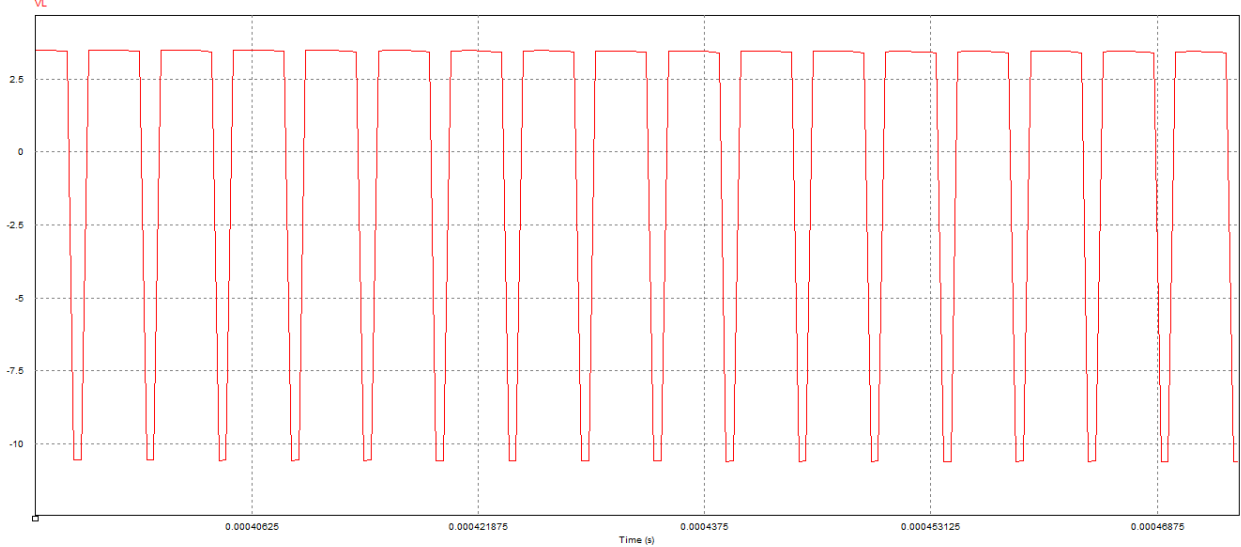
-tensiunea prin dioda:



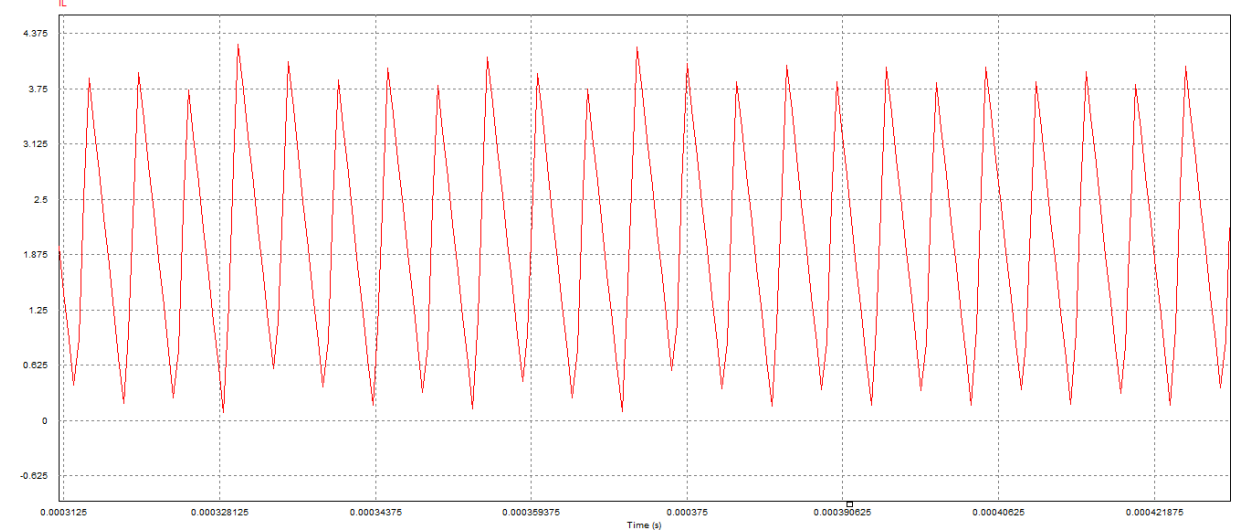
-curentul prin dioda:



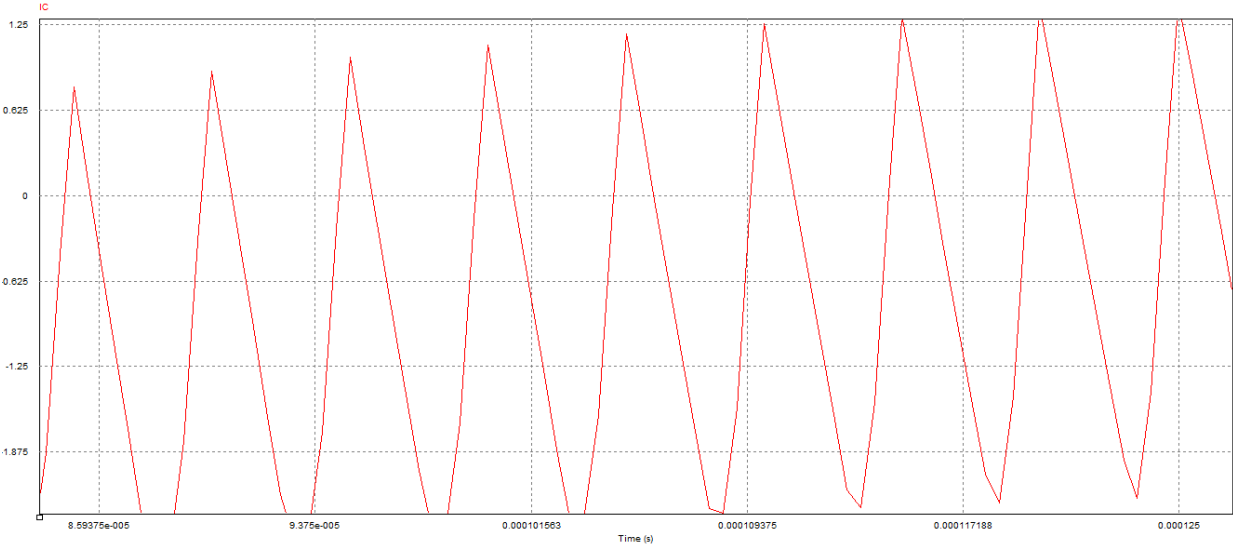
-tensiunea prin bobina:



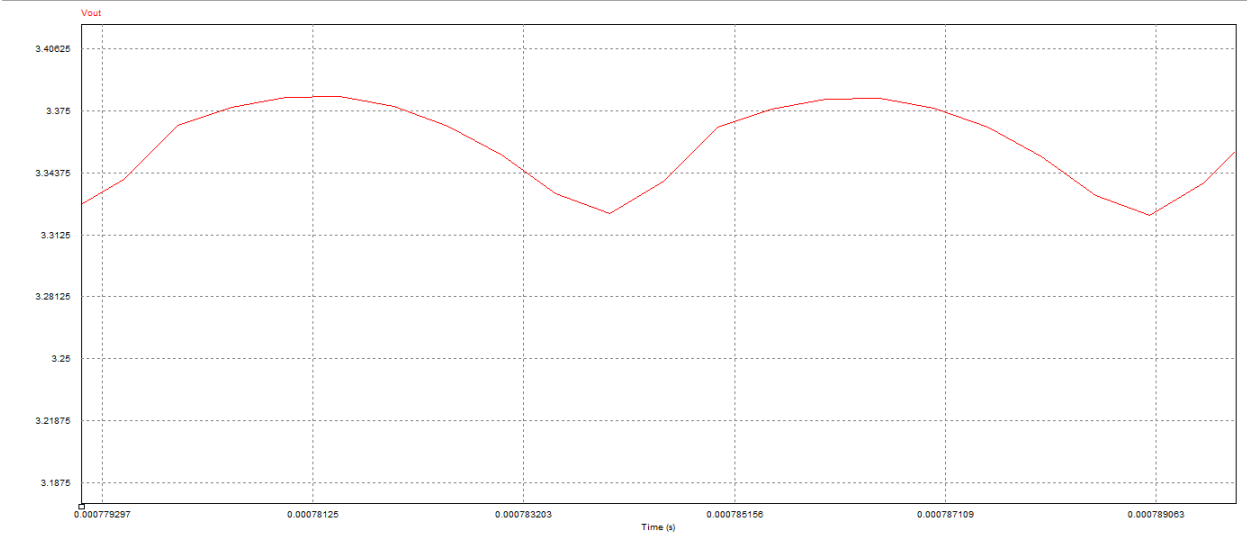
-curentul prin bobina:



-curentul prin condensator:



-tensiunea de iesire:



-curentul de iesire:

